# Automotive Technical Service Diagnostic Bulletin

**Document ID:** DIAG-C-CLASS-P0171
**Classification:** OEM Level 3 Diagnostic & Repair Procedure
**Target Vehicle:** Mercedes-Benz C-Class (W206 / C 300)
**Powertrain:** 2.0L Turbocharged Inline-4 (M254) with 2nd Gen ISG (Beltless 48V Mild-Hybrid)
**Engine Control System:** Bosch MD1CS006 (Engine Control Module N3/10) - High-Speed CAN-E2 Architecture
**Electrical Architecture:** Integrated 12V / 48V Architecture with 48V DC/DC Converter (N83/1)
**Fuel Injection Architecture:** Single Cam-Driven High Pressure Pump (HPP) with Solenoid Direct Injectors (200-250 bar)
**Firing Order:** 1-3-4-2

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## 1. DIAGNOSTIC TROUBLE CODE (DTC) SPECIFICATION

- **DTC Code:** `P0171`
- **Standard Fault Description:** System Too Lean (Bank 1)
- **System Definition:** Short-term and long-term fuel trim values on Bank 1 have exceeded the upper stoichiometric enrichment threshold (+25%).
- **MIL (Malfunction Indicator Lamp) Status:** Illuminated (Continuous or Flashing depending on catalyst severity)
- **Monitoring Condition:** Continuous monitoring during engine running cycle with coolant temp > 60°C and vehicle speed > 0 km/h.

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## 2. PLATFORM-SPECIFIC ROOT CAUSE ANALYSIS (RCA)

Root cause analysis customized specifically for the **Mercedes-Benz C-Class (W206 / C 300)**:

1. **Twin-Scroll Turbo Intake Snorkel Accordion Tear:** Flexible bellows downstream of MAF sensor cracked underneath due to engine torque movement.
2. **Brake Booster Vacuum Ejector Check Valve Failure:** Sucking unmetered atmospheric air into intake manifold past failed non-return valve.
3. **Camtronic Positive Crankcase Ventilation Valve Diaphragm Rupture:** Direct unmetered blow-by bypass into cylinder head intake tract.

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## 3. DEDICATED WIRING DIAGRAM & CIRCUIT PINOUT

Schematic configuration and electrical pin connections specifically for the C-Class control harness:

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[Manifold Absolute Pressure B28/2]         [ECM N3/10 Connector 'K']
Pin 1 (Sensor Reference GND) -------------> Pin 54 (GND)
Pin 2 (5.0V Regulated Reference) <--------- Pin 55 (5V VCC)
Pin 3 (MAP Signal 0.2 - 4.8V) ------------> Pin 56 (MAP Signal)

[Fuel Tank Vent Purge Valve Y58]
Circuit 15 (+12V Protected Fuse 15A) -----> Pin 1 (+12V)
Pin 82 (Low-Side Pulse Width Modulated) --> Pin 2 (PWM Control)
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### Electrical Reference Points

- **Primary Ground Reference:** `W10 (Engine Block Ground Point, Right) & W9 (Front Wheel Arch Ground)`
- **Power Distribution Node:** `Engine Bay Prefuse Box F32/1 & Front SAM Module (N10/1)`
- **Signal Logic Voltage Range:** 0.0V to 5.0V regulated DC (unless high-voltage direct injection rail)

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## 4. STEP-BY-STEP DIAGNOSTIC & VERIFICATION PROCEDURE

Follow this OEM workshop sequence to isolate the failure on this platform:

1. Smoke test intake system using professional smoke generator hooked to intake snorkel at 5 psi.
2. Check for smoke emerging around brake booster aspirator / vacuum lines near firewall.
3. Observe Short Term Fuel Trim (STFT) while spraying brake cleaner carefully around intake manifold gasket joints; look for STFT drop.
4. Disconnect and plug purge valve Y58 inlet line; check if STFT normalizes to 0% ± 3%.

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## 5. PRECISE RESOLUTION & REPAIR PROCEDURE

- Replace cracked turbocharger induction rubber bellows hose (Part # A254090xxxx).
- Replace vacuum ejector check valve line assembly running to brake booster.
- Perform reset of mixture adaptation (self-adaptation) via XENTRY.

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## 6. POST-REPAIR QUALITY VALIDATION & RE-TESTING

1. Clear all DTCs from Engine Control Module ( `Bosch` ) and secondary gateway modules.
2. Perform comprehensive drive cycle:
3. Cold start idle for 3 minutes (monitors secondary air / catalyst preheat and cold misfire).
4. Accelerate smoothly to 50 km/h and hold for 2 minutes.
5. Cruise at 80–100 km/h on highway for 10 minutes (validates downstream O2 sensor and fuel trim closed-loop balance).
6. Coasting deceleration (fuel cut-off check) for 5 seconds without braking.
7. Re-scan control units with diagnostic tool; confirm all OBD-II readiness monitors show **READY** / **COMPLETE** and no pending DTCs are stored.